

Scalable multiphysics and scientific AI for wave-material systems

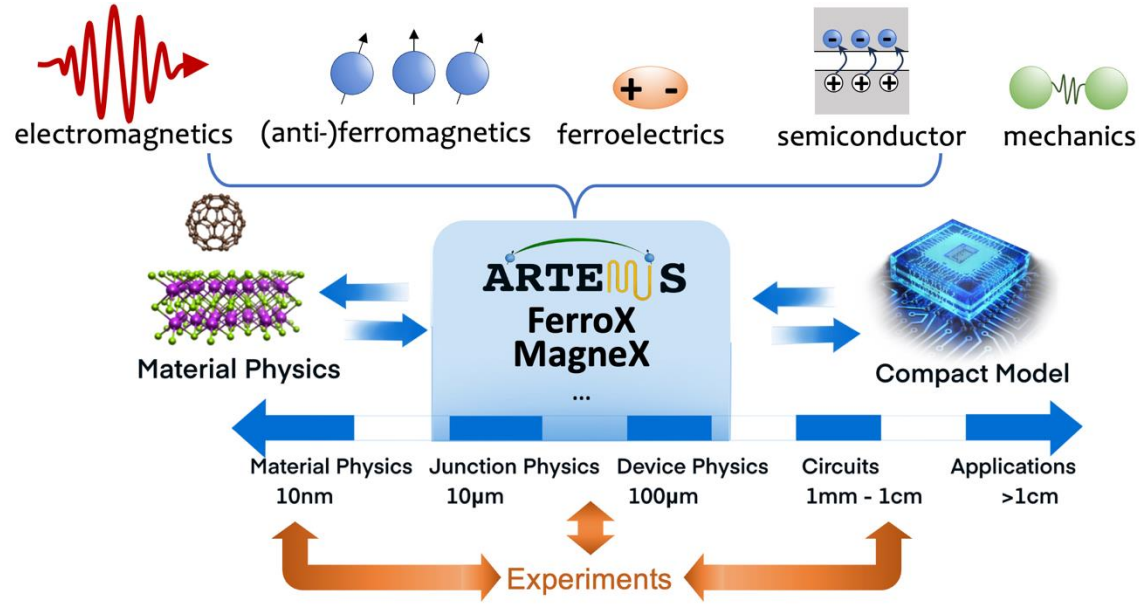
Quantum circuits, ferroic antennas and antiferromagnetic devices

Zhi (Jackie) Yao, Ramamoorthy Ramesh, Sayeef Salahuddin,
Andy Nonaka, Qi Tang, Michael Mahoney,
and many valued collaborators

Georgia Tech Electrical and Computer Engineering



MicroEleX: open GPU multiphysics [1]

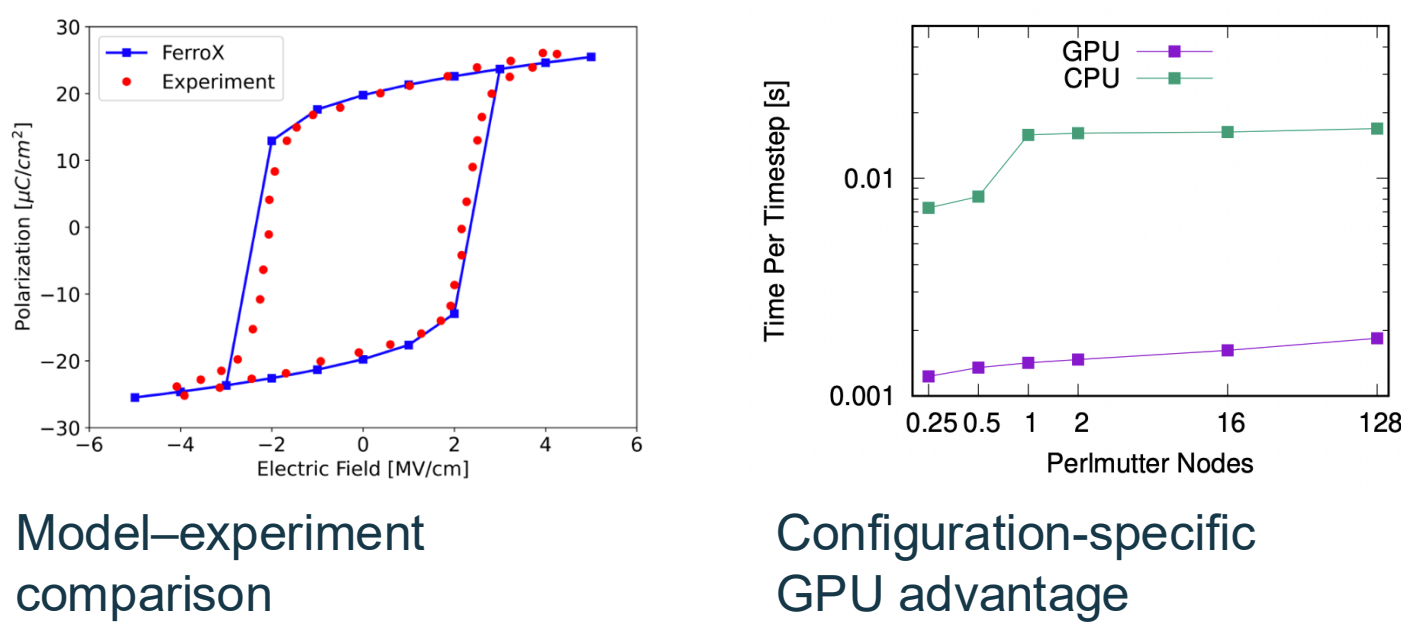


Coupled scales and nonlinear materials

Maxwell fields couple to magnetization, ferroelectric polarization and strain. Thin films, interfaces and disparate wave speeds challenge stability and cost.

AMReX-based ARTEMIS, FerroX and MagneX connect materials to device transients.

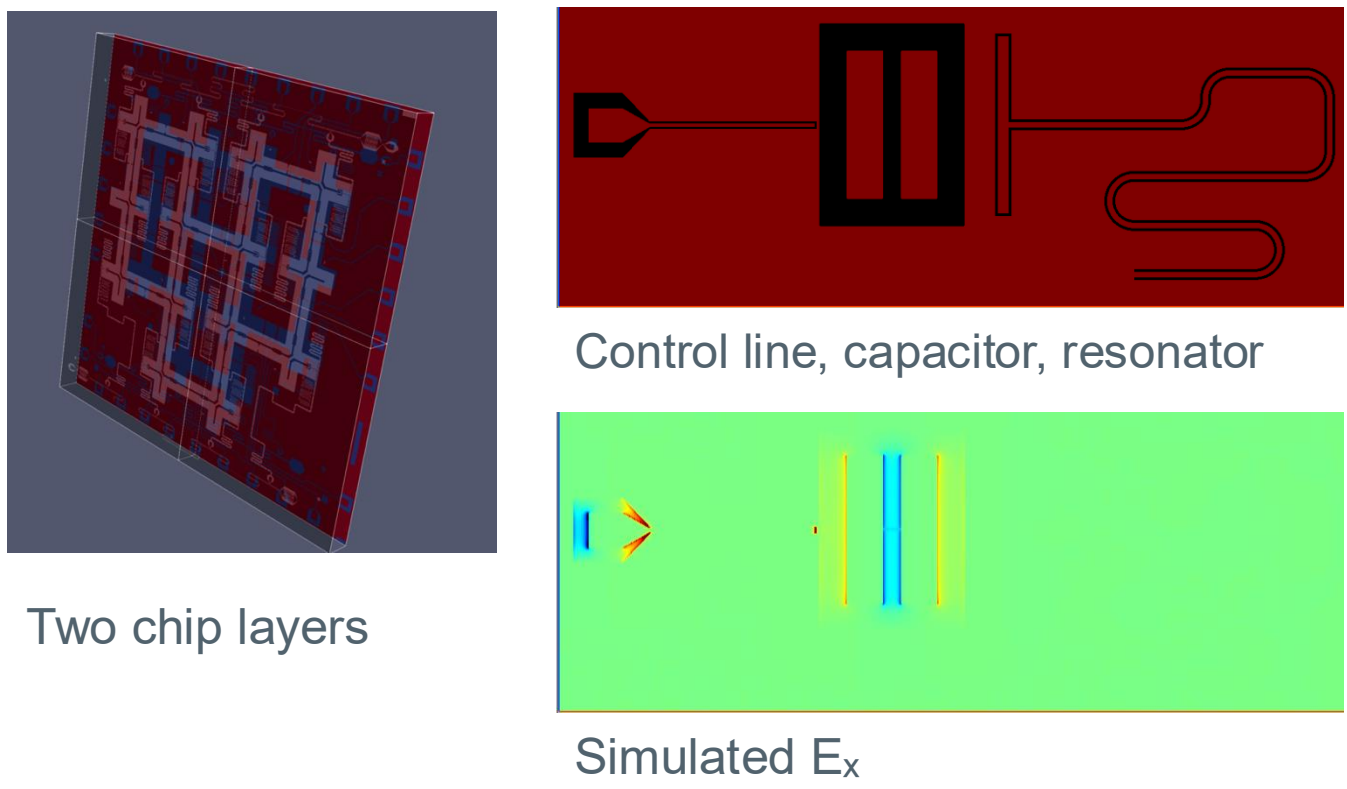
FerroX validation and GPU timing [2]



GPU-scale superconducting quantum-circuit simulations

ARTEMIS couples full-wave fields to material and circuit models. Geometry, junction nonlinearity and control-line excitation enter the transient solution, providing a physical basis for circuit-design metrics. [3]

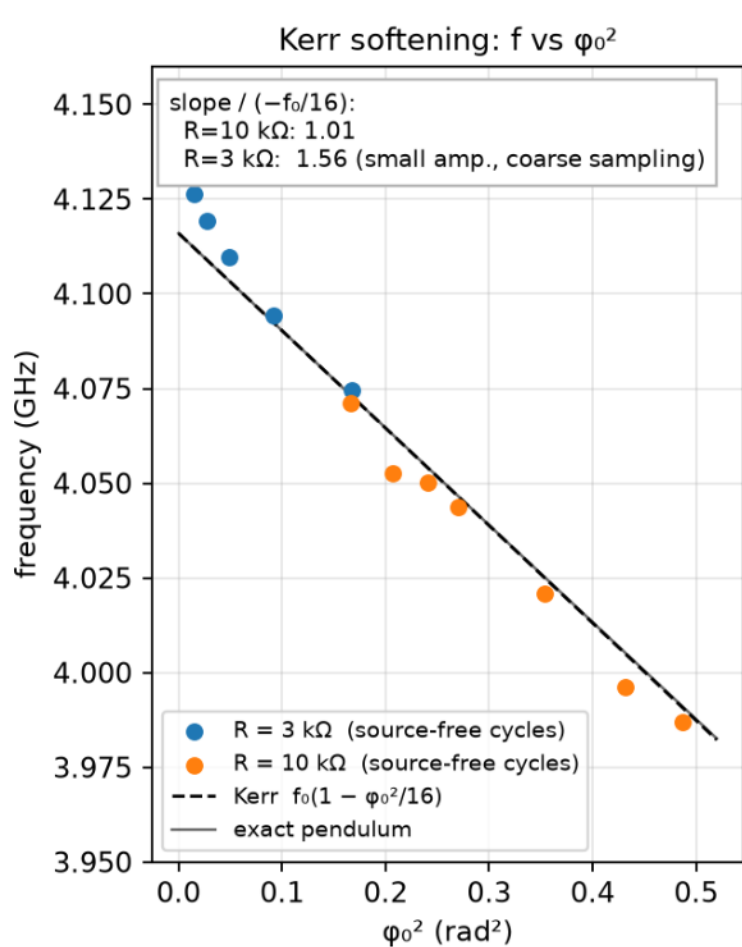
Full-chip fields and local response



11.4 billion mesh cells on 6,724 A100 GPUs

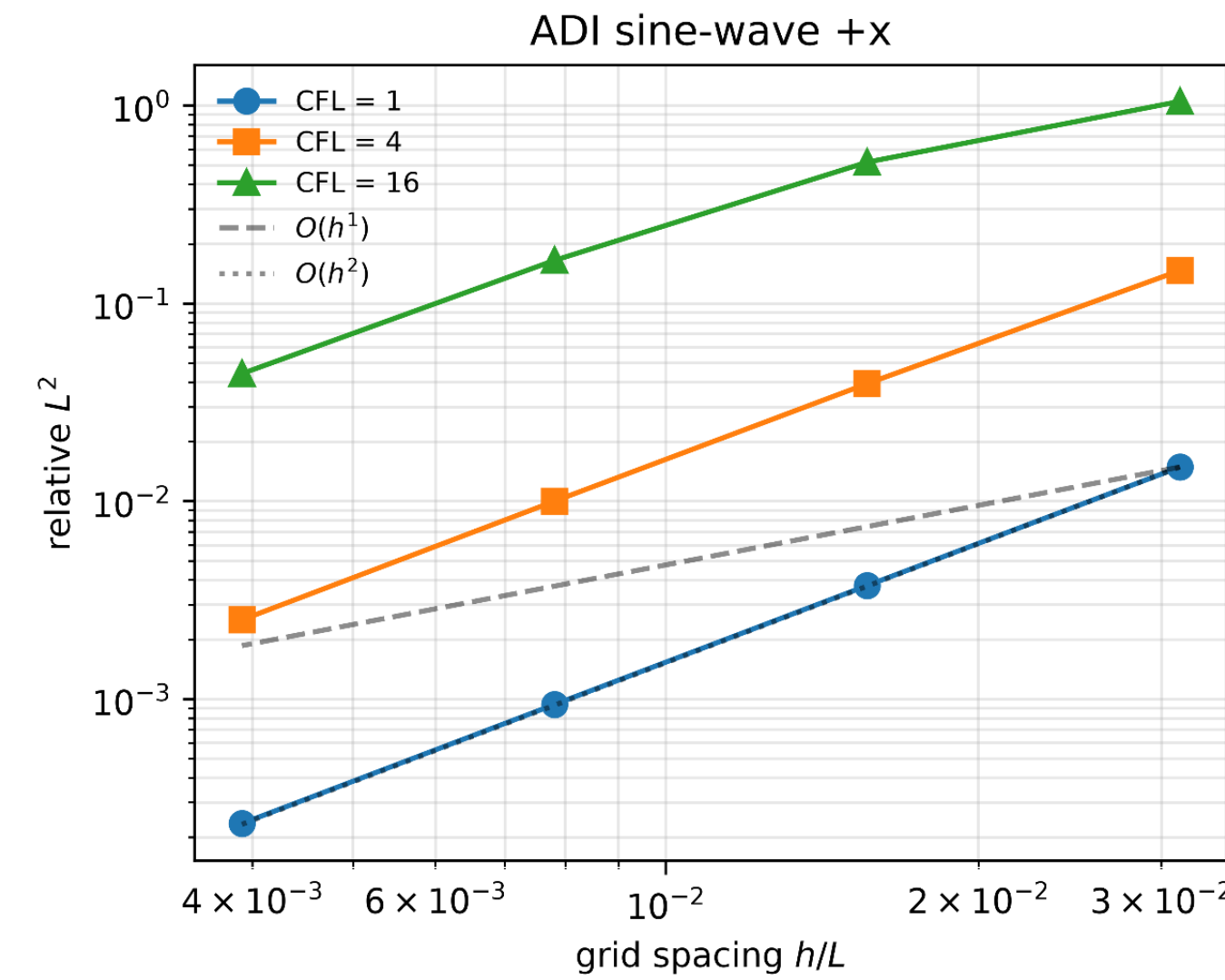
9.43 × 9.43 × 0.3 mm domain, 1.25 µm grid.
1.5 million steps follow 1 ns of chip dynamics.
Next: converged coupling and crosstalk metrics.

Nonlinear Josephson dynamics



Field-coupled RCSJ junctions reproduce amplitude-dependent frequency shifts. Shunt resistance controls ring-down damping.

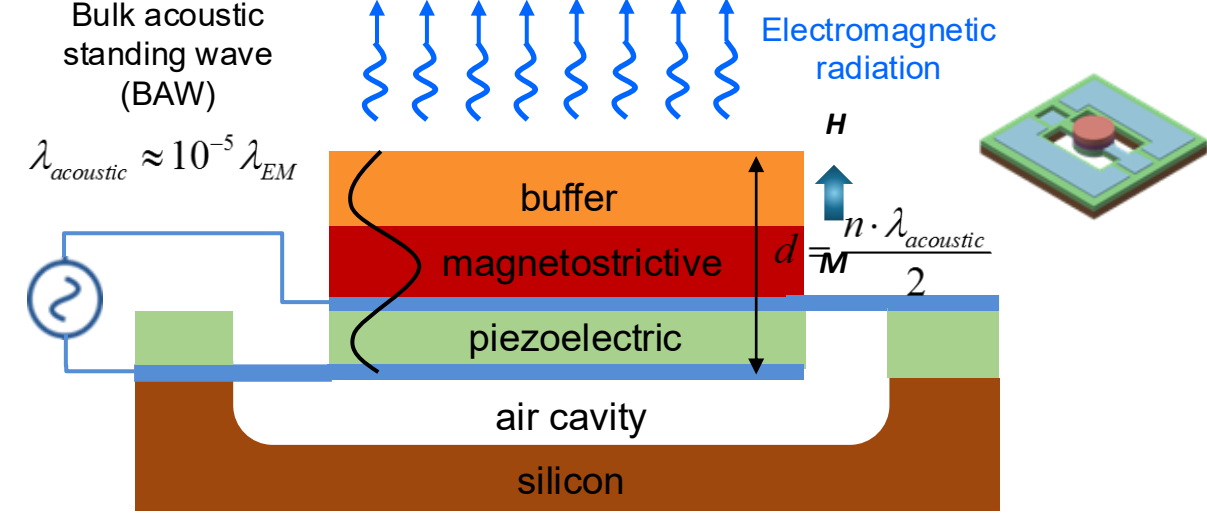
Implicit stepping with controlled error



Larger time steps reduce step count, but accuracy constrains the useful CFL increase.

Multiferroic antennas: equations, fields and device response

Voltage, strain and magnetic radiation



Applied voltage drives piezoelectric strain. Magnetoelastic coupling excites magnetization and the radiating aperture fields.

Numerical requirements

Bidirectional coupling at each time step, interface conditions and stable integration across electromagnetic and acoustic scales.

Self-consistent coupled equations [4, 5]

Maxwell

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{H} = \sigma \mathbf{E} + \frac{\partial \mathbf{D}}{\partial t}$$

Elastodynamics

$$\nabla \cdot \mathbf{T} = \rho \frac{\partial \mathbf{v}}{\partial t}, \quad \nabla_S \mathbf{v} = \frac{\partial \mathbf{S}}{\partial t}$$

Magnetization dynamics (LLG)

$$\frac{\partial \mathbf{M}}{\partial t} = \mu_0 \gamma (\mathbf{M} \times \mathbf{H}_{\text{eff}}) - \frac{\alpha}{|\mathbf{M}|} \mathbf{M} \times \frac{\partial \mathbf{M}}{\partial t}$$

Piezoelectric and magnetostrictive constitutive laws

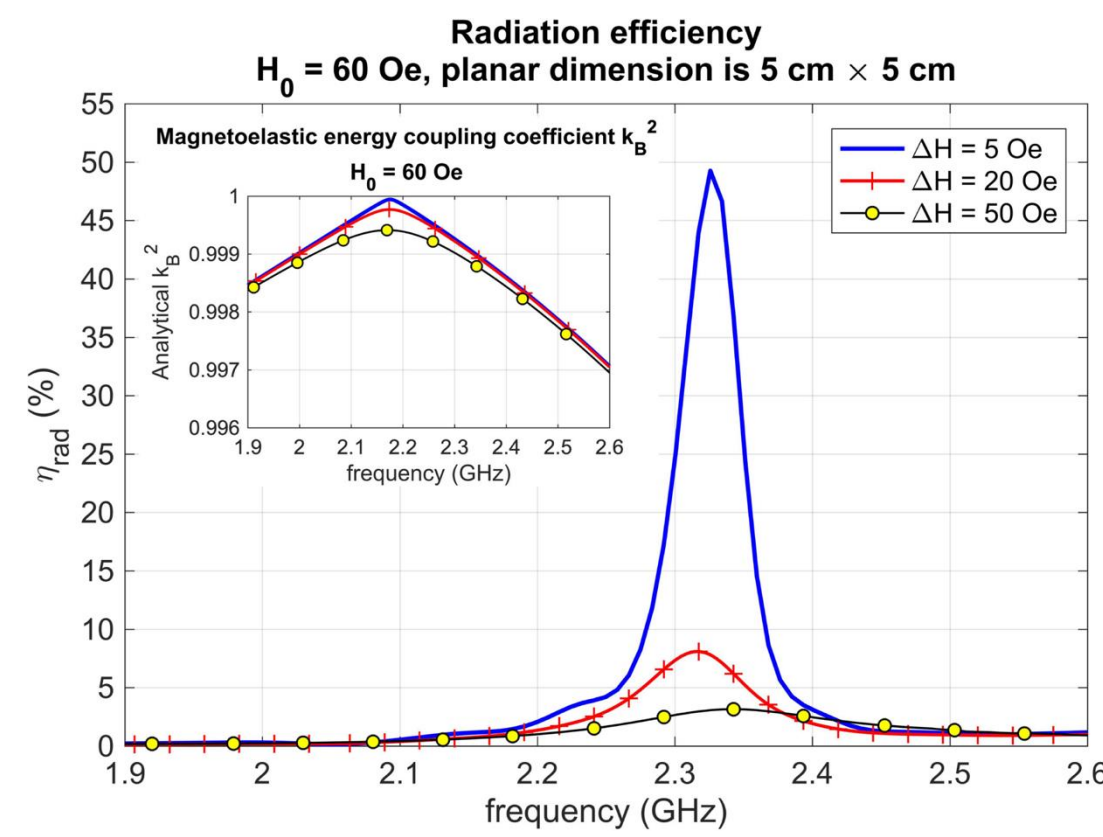
$$\mathbf{D} = \epsilon^T \mathbf{E} + d_E \mathbf{T}$$

$$\mathbf{B} = \frac{\chi^T + 1}{\chi^T} \mu_0 \mathbf{M} - \mu_0 d_B \mathbf{T}$$

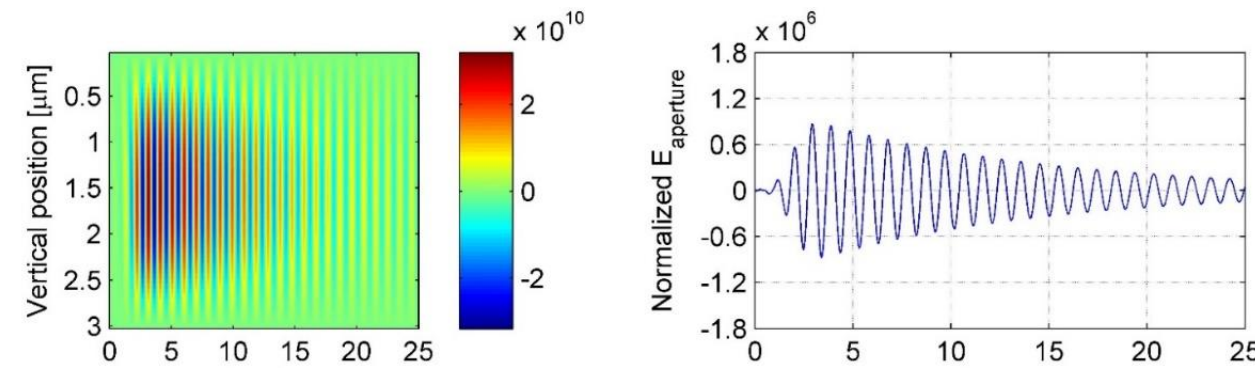
$$\mathbf{S} = d_E \mathbf{E} + s^E \mathbf{T} - \eta s^E \frac{\partial \mathbf{S}}{\partial t}$$

$$\mathbf{S} = s^B \mathbf{T} - \eta s^B \frac{\partial \mathbf{S}}{\partial t} + d_B \cdot \mu_0$$

Simulated efficiency and acoustic modes

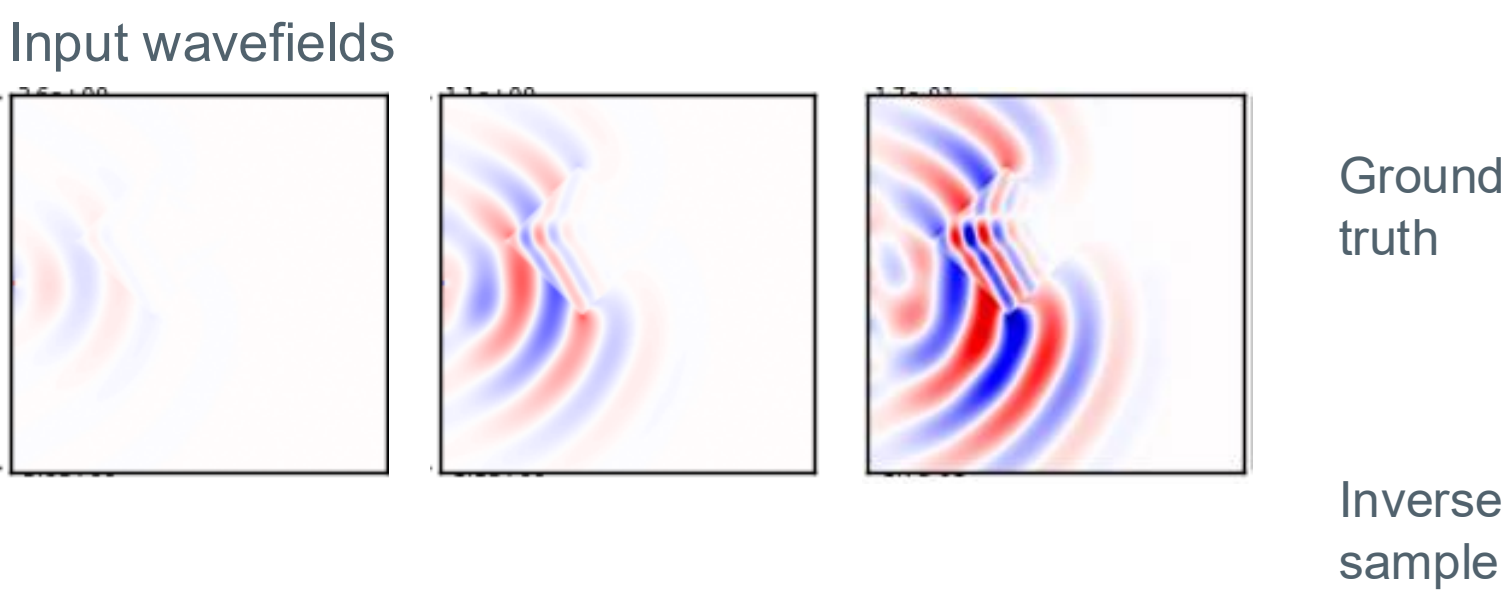


5 × 5 cm aperture; bias H₀ = 60 Oe. Resonance alignment and damping set efficiency.



AI for wave physics

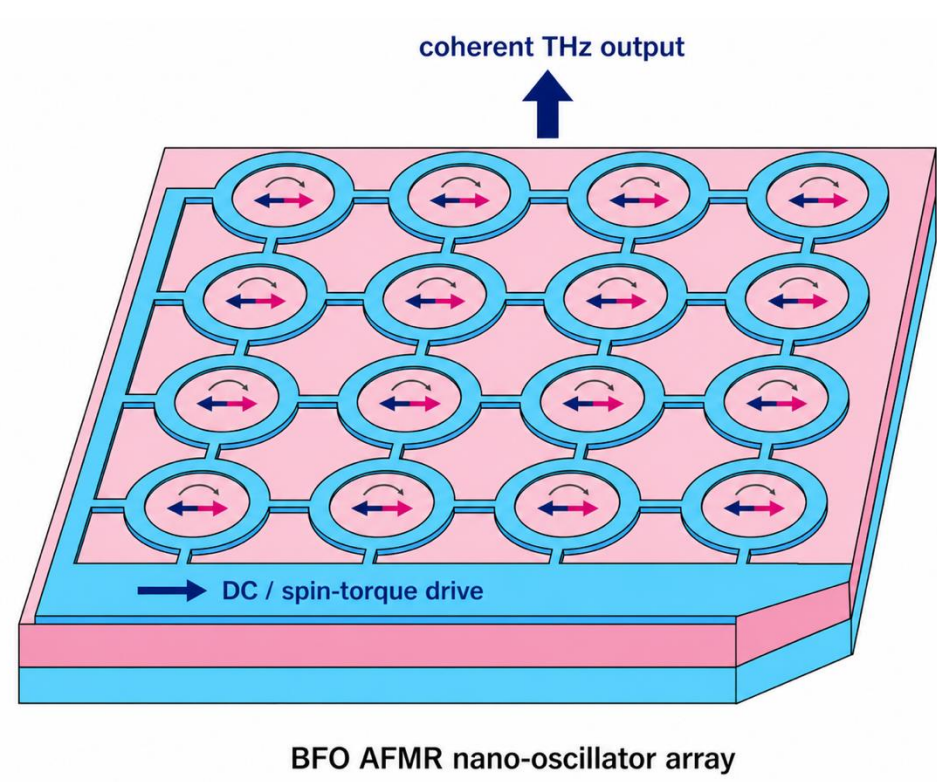
WaveChat: forward, inverse and language tasks



A shared spatiotemporal model predicts future fields and reconstructs material structure from wavefields. Ongoing work: test unfamiliar geometries, quantify uncertainty and enforce physical consistency.

AFM nano-oscillators at THz

Proposed BFO array: advanced simulation needed



Two-sublattice dynamics

Resolve exchange, anisotropy and coupling to polarization.

Driven nonlinear response

Predict threshold, amplitude stability and thermal effects.

Coupled arrays

Test synchronization, phase noise and output loading.

An active drive must compensate AFMR loss. Drive mechanism and useful output remain to be demonstrated.

Seeking MFEM collaborators

High-order / adaptive FEM for thin films and chip geometry. Energy-consistent coupling, implicit solvers and GPU preconditioning.

[1] IEDM (2025). [2] CPC 290,108757 (2023). [3] IEEE Quantum Week (2026). [4] IEEE TAP 63,3335–3344 (2015). [5] JMMCT 5 (2019); IJHPCA 36 (2022).

Code: github.com/AMReX-Microelectronics

Let's define a shared benchmark

We bring open codes and device physics. Compare convergence, energy balance and cost.

zhijackeyao@ece.gatech.edu